# Supplementary material: Viability of keV sterile-neutrino dark matter under BBN and X-ray constraints, 2024–2026

HEP-Genesis-Agent session record
Project: ramp-combo-sterile-bbn-xray

2026-10-21

*Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.*

## A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

### A.1 Session

- Source: cli client; conversation `1060b458`
- Started: 2026-10-21 14:31:00; report generated: 2026-10-21 14:43:29
- Project: ramp-combo-sterile-bbn-xray (`/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray`)
- Compute execution site(s) selected in the session: local
- 1 user turn(s), 103 tool call(s) (0 returned errors), 2 figure(s), 11 data file(s)

### A.2 Models

| Stage | Model | Endpoint |
|---|---|---|
| quick | `claude-opus-5-5` | `https://mag.genesis.americansciencecloud.org/v1` |
| report | `claude-opus-5-5` | `https://mag.genesis.americansciencecloud.org/v1` |
| coder | `claude-opus-5-5` | `https://mag.genesis.americansciencecloud.org/v1` |

### A.3 MCP servers

| Server | Transport | Location |
|---|---|---|
| hep-research-aid | stdio | `python -m hep_genesis.domain.mcp_server` |

| Server | Transport | Location |
|---|---|---|
| alcf-tools | stdio | `python -m hep_genesis.iri.alcf.mcp_server` |
| nersc-tools | stdio | `python -m hep_genesis.iri.nersc.mcp_server` |

### A.4 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 1 | `serve-sbbn.sh list_sterile_nu_skills` | `{}` | `{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...` |
| 2 | `serve-hydro.sh list_observables` | `{}` | `{ "status": "success", "message": "14 observables available: GSMF, HMF, fGas, Pk-ratio, CSFR, CGD, CGED, CPP, CTP, CEP, CEEP, CMP, CYP, Pk_GO.", "files": [], "metadata": { "observables": {...` |
| 3 | `hep-research-aid list_agent_files` | `{}` | `[]` |
| 4 | `hep-research-aid list_input_files` | `{}` | `{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }` |
| 5 | `hep-research-aid list_datasets` | `{}` | `{'count': 0, 'datasets': []}` |
| 6 | `hep-research-aid list_files` | `{}` | `Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - w...` |
| 7 | `hep-research-aid get_dispatch` | `{}` | `Dispatch: local execution.` |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 8 | local search_arxiv_ abstracts | {"query": "sterile neutrino dark matter X-ray constraints 3.5 keV line 2024", "max_results": 8} | [1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of... |
| 9 | local search_arxiv_ abstracts | {"query": "sterile neutrino big bang nucleosynthesis helium deuterium constraints lepton asymmetry", "max_ results": 8} | [2005.06721v2] Big Bang Nucleosynthesis constraints on sterile neutrino and lepton asymmetry of the Universe Abstract: We consider the cosmological effects of sterile neutrinos with the masses of $150... |
| 10 | local search_arxiv_ abstracts | {"query": "sterile neutrino dark matter parameter space Shi-Fuller resonant production constraints Lyman-alpha", "max_results": 8} | [1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of... |
| 11 | hep-research-aid search_arxiv | {"query": "sterile neutrino dark matter X-ray NuSTAR XRISM eROSITA constraints", "max_results": 10, "sort_ by": "submittedDate"} | Found 10 papers matching query: 'sterile neutrino dark matter X-ray NuSTAR XRISM eROSITA constraints' (source: arXiv API) Paper 1: arXiv ID: 2610.28633v1 Title: Constraints on Millimeter-wave Extraga... |
| 12 | local search_arxiv_ abstracts | {"max_results": 8, "query": "large lepton asymmetry BBN consistent sterile neutrino dark matter 2025"} | [1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 13 | local search_arxiv_ abstracts | {"max_results": 6, "query": "XRISM 3.5 keV line search Perseus dark matter decay"} | [2505.07479v2] Thermal and Kinematic Properties of Ejecta in SN1987A revealed by XRISM Abstract: We present an analysis of high-resolution spectra from the shock-heated plasmas in SN~1987A, based on a... |
| 14 | local get_arxiv_ metadata | {"arxiv_id": "2507.20659"} | arXiv ID: 2507.20659 Title: Maximal parameter space of sterile neutrino dark matter with lepton asymmetries Authors: Kensuke Akita, Koichi Hamaguchi, Maksym Ovchynnikov Year: 2025 Abstract: We delinea... |
| 15 | local get_arxiv_ metadata | {"arxiv_id": "2507.18752"} | arXiv ID: 2507.18752 Title: Return of the Lepton Number: Sterile Neutrino Dark Matter Production and the Revival of the Shi-Fuller Mechanism Authors: Cannon M. Vogel, Helena García Escudero, Julien Fr... |
| 16 | local get_arxiv_ metadata | {"arxiv_id": "2510.24560"} | arXiv ID: 2510.24560 Title: XRISM constraints on unidentified X-ray emission lines, including the 3.5 keV line, in the stacked spectrum of ten galaxy clusters Authors: XRISM Collaboration, Marc Audar... |
| 17 | local search_arxiv_ abstracts | {"max_results": 6, "query": "eROSITA or NuSTAR decaying dark matter sterile neutrino mixing angle limit 2024"} | [1606.02684v1] Neutrino Oscillations With Two Sterile Neutrinos Abstract: This work estimates the probability of $\mu$ to $e$ neutrino oscillation with two sterile neutrinos using a 5x5 U-matrix, an ext... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 18 | local search_arxiv_ abstracts | {"max_results": 5, "query": "lepton flavor asymmetries washed out neutrino oscillations BBN CMB constraints large asymmetry Froustey"} | [9906311v2] Detailed study of BBN implications of neutrino oscillation generated neutrino asymmetries in some four neutrino models Abstract: We re-examine the evolution of neutrino asymmetries in se... |
| 19 | local get_arxiv_ metadata | {"arxiv_id": "2502.17374"} | arXiv ID: 2502.17374 Title: Playing with lepton asymmetry at the resonant production of sterile neutrino dark matter Authors: Dmitry Gorbunov, Dmitry Kalashnikov, George Krugan Year: 2025 Abstract: We... |
| 20 | serve-sbbn.sh describe_sterile_ nu_tools | {} | { "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n= ========== ==========\n\ nMaps sterile-neutrino particle parameters (m_s, sin^2 2... |
| 21 | serve-sbbn.sh list_sterile_nu_ skills | {} | { "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ... |
| 22 | serve-sbbn.sh load_sterile_nu_ skill | {"name": "kev-sterile-dm-xray"} | { "status": "ok", "files": [], "message": "Loaded skill 'kev-sterile-dm-xray'. Follow these instructions.", "metadata": { "instructions": "--\nname: kev-sterile-dm-xray\ ndescription: keV ... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 23 | serve-sbbn.sh scan_sterile_nu_parameters | {"m_min_eV": 1000, "m_max_eV": 100000, "theta_min": 1e-14, "theta_max": 1e-06, "n_points": 60, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100"} | { "status": "ok", "files": [ "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/scan_DW_9bfc3d.csv", "/Users/nramachandra/hep-genesis-testing/... |
| 24 | local download_arxiv_full_text | {"arxiv_id": "2405.06509", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100"} | Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2405.06509.txt |
| 25 | local download_arxiv_full_text | {"arxiv_id": "2510.24560", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100"} | Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2510.24560.txt |
| 26 | local download_arxiv_full_text | {"arxiv_id": "2507.18752", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100"} | Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2507.18752.txt |
| 27 | serve-sbbn.sh predict_sterile_nu_grid | {"masses_eV": [7100, 10000, 20000, 50000], "sin2_2thetas": [1e-14, 1e-13, 1e-12, 3e-12, 1e-11, 3e-11, 7e-11, 1e-10, 2e-10, 3e-10, 5e-10, 1e-09, 2e-09, 3e-09, 5e-09, 1e-08, 3e-08, 1e-07, 3e-07, 1e-06], "output_dir": "/Use... | { "status": "ok", "files": [ "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/sterile_nu_grid_DW_097459.csv" ], "message": "Predicted 80 poi... |
| 28 | local download_arxiv_full_text | {"arxiv_id": "1609.00667", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100"} | Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/1609.00667.txt |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 29 | serve-sbbn.sh predict_sterile_nu_xray_line | {"m_s_eV": 7100, "sin2_2theta": 4e-10, "target": "all"} | { "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 7.1 keV, sin\u00b22\u03b8 = 4e-10\n\nE_\u03b3 = 3.55 keV\nDecay... |
| 30 | hep-research-aid read_text_file | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/sterile_nu_grid_DW_097459.csv"} | m_s_eV,sin2_2theta, delta_neff,neff,omega_s_h2,Yp,DH_x1e5,Li7H_x1e10,chi2_bbn,delta_chi2_bbn,lifetime_s, free_streaming_kpc,bbn_excluded_95,overcloses, bbn_treatment_valid 7100, 1e-14,6.18479e-08,3.044, 4.... |
| 31 | local download_arxiv_full_text | {"arxiv_id": "2507.20659", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100"} | Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2507.20659.txt |
| 32 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2510.24560.txt", "length": 12000} | [chars 0-12000 of 67438] (more available, next start=12000) Title: XRISM constraints on unidentified X-ray emission lines, including the 3.5 keV line, in the stacked spectrum of ten galaxy clusters A... |
| 33 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/1609.00667.txt", "length": 12000} | [chars 0-12000 of 51891] (more available, next start=12000) Title: (Almost) Closing the $\nu$MSM Sterile Neutrino Dark Matter Window with NuSTAR Authors: Kerstin Perez, Kenny C. Y. Ng, John F. Beacom, ... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 34 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2405.06509.txt", "length": 6000} | [chars 0-6000 of 100042] (more available, next start=6000) Title: Constraints on primordial lepton asymmetries with full neutrino transport Authors: Julien Froustey, Cyril Pitrou Year: 2024 arXiv ID:... |
| 35 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2507.18752.txt", "length": 8000} | [chars 0-8000 of 62265] (more available, next start=8000) Title: Return of the Lepton Number: Sterile Neutrino Dark Matter Production and the Revival of the Shi-Fuller Mechanism Authors: Cannon M. Vo... |
| 36 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2507.20659.txt", "length": 6000} | [chars 0-6000 of 145575] (more available, next start=6000) Title: Maximal parameter space of sterile neutrino dark matter with lepton asymmetries Authors: Kensuke Akita, Koichi Hamaguchi, Maksym Ovch... |
| 37 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2510.24560.txt", "start": 12000, "length": 14000} | [chars 12000-26000 of 67438] (more available, next start=26000) Cambridge, MA 02138, USA 70School of Physics & Astronomy, University of Nottingham, University Park, Nottingham NG7 2RD, UK -- Page ... |
| 38 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/1609.00667.txt", "start": 20000, "length": 14000} | [chars 20000-34000 of 51891] (more available, next start=34000) es, is multiplied by the energy-dependent e?ciency for photons to pass through the detector beryllium shield. All model components incl... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 39 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2507.20659.txt", "start": 6000, "length": 7000} | [chars 6000-13000 of 145575] (more available, next start=13000) 101 102 ms [keV] 10-16 10-14 10-12 10-10 10-8 sin2 2$\theta$ overproduction underproduction no asymmetry Lμ = -L$\tau$ = 0.06 Lμ = -L$\tau$ = 0.018 X-r... |
| 40 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2507.18752.txt", "start": 36000, "length": 12000} | [chars 36000-48000 of 62265] (more available, next start=48000) n the top panel of Fig. 5, so that the asym- metries are small enough that the structure formation is mostly sensitive to the sterile n... |
| 41 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2510.24560.txt", "length": 16000, "start": 36000} | [chars 36000-52000 of 67438] (more available, next start=52000) d combined all 36 of them into the final NXB model for the de-redshifted stacked spectrum. The NXB model is used with a di- -- Page 9... |
| 42 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/1609.00667.txt", "length": 10000, "start": 34000} | [chars 34000-44000 of 51891] (more available, next start=44000) the galaxy counting constraint ob- tained by Ref. [71], whose limit is comparable to that in Ref. [70]. Both analyses take into accoun... |
| 43 | serve-sbbn.sh predict_sterile_nu_grid | {"masses_eV": [7100, 10000, 20000, 50000], "sin2_2thetas": [2.545e-10, 1.374e-10, 3.946e-11, 7.583e-12], "combine": "zip", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/... | { "status": "ok", "files": [ "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/sterile_nu_grid_DW_1b6ff9.csv" ], "message": "Predicted 4 poin... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 44 | serve-sbbn.sh predict_sterile_nu_xray_line | {"m_s_eV": 7100, "sin2_2theta": 2.545e-10} | { "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 7.1 keV, sin\u00b22\u03b8 = 2.5e-10\n\nE_\u03b3 = 3.55 keV\nDec... |
| 45 | serve-sbbn.sh predict_sterile_nu_xray_line | {"m_s_eV": 10000, "sin2_2theta": 1.374e-10} | { "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 10 keV, sin\u00b22\u03b8 = 1.4e-10\n\nE_\u03b3 = 5 keV\nDecay r... |
| 46 | serve-sbbn.sh predict_sterile_nu_xray_line | {"m_s_eV": 20000, "sin2_2theta": 3.946e-11} | { "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 20 keV, sin\u00b22\u03b8 = 3.9e-11\n\nE_\u03b3 = 10 keV\nDecay ... |
| 47 | serve-sbbn.sh predict_sterile_nu_xray_line | {"m_s_eV": 50000, "sin2_2theta": 7.583e-12} | { "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 50 keV, sin\u00b22\u03b8 = 7.6e-12\n\nE_\u03b3 = 25 keV\nDecay ... |
| 48 | serve-sbbn.sh predict_sterile_nu | {"m_s_eV": 7100, "sin2_2theta": 7e-11, "mechanism": "SF", "lepton_asymmetry": 0.001} | { "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 7.1e+03 eV, sin\u00b22\u03b8 = 7e-11\nMechanism: SF (SF Delta N_eff is a heuristic (0.1 x the DW value), not a resonant-product... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 49 | board post_finding | {"claim": "Vogel et al. (2507.18752): Shi-Fuller sterile-nu DM is viable for initial L >~ 0.5 (L given at T=10 GeV), m_s > ~ 35 keV and sin^2 2theta < ~ 1e-14 once X-ray limits (NuSTAR+INTEGRAL/SPI, 95% CL) and the structu... | Recorded finding f-001. |
| 50 | board post_finding | {"claim": "2507.18752 provides fitting functions valid for 15 keV <= m_s <= 60 keV and 1e-15 <= sin^2 2theta <= 3e-13: m_th = 1.329 m_s^1.044 (sin^2 2theta)^0.055 (keV; average error 1.3%) and the L needed for all DM, L ... | Recorded finding f-002. |
| 51 | board post_finding | {"claim": "Akita, Hamaguchi, Ovchynnikov (2507.20659): with large flavor asymmetries and zero total asymmetry (washed out by oscillations at T_osc ~ 15 MeV), the viable sin^2 2theta range grows by up to two orders of mag... | Recorded finding f-003. |
| 52 | board post_finding | {"claim": "Model dependence: Froustey and Pitrou (2405.06509), using full 3-flavor neutrino transport plus BBN, find that 'the space of allowed initial asymmetries is generically unbound' with current abundance and CMB d... | Recorded finding f-004. |
| 53 | board post_finding | {"claim": "Sterile-nu tools: BBN uses a PRyMordial fit (Yp, D/H, Li7) driven only by Delta N_eff, with exclusion at Delta chi^2 > 3.84 against Yp = 0.2449 +/- 0.004 (Aver+ 2021), D/H = 2.547 +/- 0.025 e-5 (Cooke+ 2018) a... | Recorded finding f-005. |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 54 | board post_finding | {"claim": "DW scan (m_ s 1-100 keV, sin^2 2theta 1e-14 to 1e-6, 60x60): 546 of 3600 points are BBN-excluded and 1740 overclose. The BBN 95% upper limit on sin^2 2theta is 3.9e-7 at 1 keV, 8.2e-8 at 7 keV, 4.4e-8 at 13 keV... | Recorded finding f-006. |
| 55 | board post_finding | {"claim": "Tool BBN numbers (DW, 84 points): the SM baseline is Yp = 0.246815, D/H = 2.4433e-5 and N_eff = 3.044. This is +0.48 sigma in Yp and -4.1 sigma in D/ H from the observed values (chi^2 = 2.77). On the DW relic c... | Recorded finding f-007. |
| 56 | board post_finding | {"claim": "X-ray decay on the DW relic curve: Gamma = 6.25e-28 s^-1 (7.1 keV, E = 3.55 keV), 1.87e-27 (10 keV), 1.72e-26 (20 keV) and 3.23e-25 (50 keV). These agree within about 1.5% with the analytic 1.38e-29 (sin^2 2th... | Recorded finding f-008. |
| 57 | serve-sbbn.sh plot_sterile_nu_ exclusion | {"scan_file": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-sterile-bbn-xray/ runs/261021_143100/scan_ DW_9bfc3d.npz", "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-sterile-bbn-x... | { "status": "ok", "files": [ "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-sterile-bbn-xray/ runs/261021_143100/ exclusion_scan_DW_ 9bfc3d.png" ], "message": "Exclusion plot sa... [Fig. 10] |
| 58 | serve-sbbn.sh predict_sterile_ nu | {"m_s_eV": 37000, "sin2_ 2theta": 1.5e-14, "mechanism": "SF", "lepton_ asymmetry": 0.1} | { "status": "ok", "files": [], "message": "Sterile neutrino: m_ s = 3.7e+04 eV, sin\ u00b22\u03b8 = 1.5e-14\ nMechanism: SF (SF Delta N_eff is a heuristic (0.1 x the DW value), not a resonant-produ... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 59 | local cite | {"arxiv_id": "2507.18752", "note": "[\"Interestingly, if one adopts the Zelko+ bounds, the remaining unconstrained parameter space requires an\", \ "window for SF to account for the entirety of the dark mat-\"]", "bib_pat... | Cited: Return of the Lepton Number: Sterile Neutrino Dark Matter Production and the Revival of the Shi-Fuller Mechanism (\cite{arxiv250718752}) |
| 60 | local cite | {"arxiv_id": "2507.20659", "note": "[\"For sterile neutrino masses up to 60 keV, the viable range of mixing angles extends by up to two orders of magnitude\"]", "bib_path": "/Users/nramachandra/ hep-genesis-testing/projec... | Cited: Maximal parameter space of sterile neutrino dark matter with lepton asymmetries (\ cite{arxiv250720659}) |
| 61 | local cite | {"arxiv_id": "2405.06509", "note": "[\"Overall, we find that the space of allowed initial asymmetries is generically unbound when\"]", "bib_path": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-sterile-bbn-... | Cited: Constraints on primordial lepton asymmetries with full neutrino transport (\ cite{arxiv240506509}) |
| 62 | board post_finding | {"claim": "XRISM 2510.24560 (stacked spectra of 10 clusters, 3.75 Ms) gives a 3-sigma upper limit of 0.95e-6 ph/cm^2/s on line flux at the 3.55 keV line, which converts to a decay-rate limit of Gamma < 0.97e-27 s^-1 for ... | Recorded finding f-009. |
| 63 | board post_finding | {"claim": "The tool's X-ray decay rate at m_s = 7.1 keV with sin^2 2theta = 4e-10 is 9.82e-28 s^-1 (lifetime 1.02e27 s). This agrees to within 2% with the XRISM limit of 0.97e-27 s^-1, so the tool's rate normalization ma... | Recorded finding f-010. |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 64 | board post_finding | {"claim": "The NuSTAR Galactic Center paper (1609.00667) gives its sterile-neutrino line flux as F of about 2.6e-6 cm^ -2 s^-1 (m/20 keV)^4 (sin^ 2 2theta/1e-14)(DeltaOmega/ 4 deg^2)(J/40 GeV cm^-3 kpc sr^-1), using the cor... | Recorded finding f-011. |
| 65 | board post_finding | {"claim": "I cross-checked the tool against the 2507.18752 benchmark point, with SF production at m_ s = 37 keV, sin^2 2theta = 1.5e-14 and L = 0.1 (the tool maximum). The tool gives Delta N_eff = 3.5e-8, Yp = 0.24682, D/... | Recorded finding f-012. |
| 66 | local cite | {"arxiv_id": "2510.24560", "note": "[\"which corresponds to a DM decay rate of 0.97×10 -27\", \"sin22$\theta$/10-10 <4.1(E/ 3.55) -4f1\"]", "bib_path": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-sterile-bbn-xra... | Cited: XRISM constraints on unidentified X-ray emission lines, including the 3.5 keV line, in the stacked spectrum of ten galaxy clusters (\ cite{arxiv251024560}) |
| 67 | local cite | {"arxiv_id": "1609.00667", "note": "[\"Near$m_\chi$ = 20 keV, our result improves the limit by about\", \ "limit translates into an upper limit on the sterile neu-\"]", "bib_path": "/Users/nramachandra/ hep-genesis-testing/proje... | Cited: (Almost) Closing the $\nu$MSM Sterile Neutrino Dark Matter Window with NuSTAR (\ cite{arxiv160900667}) |
| 68 | serve-sbbn.sh predict_sterile_ nu_xray_line | {"m_s_eV": 20000, "sin2_ 2theta": 1e-12, "target": "Milky_Way_center"} | { "status": "ok", "files": [], "message": "X-ray line prediction for Milky_Way_center\nm_ s = 20 keV, sin\u00b22\ u03b8 = 1e-12\n\nE_\u03b3 = 10 keV\nDecay rate = 4.36e-28 s\u207b\u00b9\ nLifetime ... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 69 | board post_finding | {"claim": "DW exclusion-plane plot for m_s 1-100 keV, sin^2 2theta 1e-14 to 1e-6 (Figure 10, exclusion_scan_DW_9bfc3d.png). Viable DW region: the Omega h^2 = 0.12 line runs from sin^2 2theta = 2.5e-10 at 7.1 keV to 7.6e-... | Recorded finding f-013. |
| 70 | board post_dead_end | {"what": "Using the tool's SF (Shi-Fuller) mode with lepton_asymmetry to compute resonant relic density and BBN.", "why": "The tool labels SF Delta N_eff as a heuristic (0.1 x DW). It returned Omega h^2 = 432 at 7.1 keV,... | Recorded dead end x-01. |
| 71 | board post_question | {"text": "The tool's SM BBN baseline gives D/H = 2.443e-5, which is -4.1 sigma from Cooke+ 2.547 +/- 0.025e-5. Is this a calibration or nuclear-rate choice, and does it bias the Delta chi^2 exclusion contours?"} | Recorded question q-05. |
| 72 | board post_finding | {"claim": "Table of upper limits on sin^2 2theta from X-rays. From XRISM 2510.24560 (3-sigma limit, my arithmetic on the paper's polynomial): m_s = 6 keV, about 1.3e-9; 7.1 keV, about 4-5e-10 (3.9e-10 from Gamma < 0.97e-... | Recorded finding f-014. |
| 73 | hep-research-aid get_paper_figures | {"paper_id": "1609.00667", "figures": "7,8", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100"} | {"status": "error", "files": [], "message": "No usable HTML rendering for arXiv:1609.00667 (arxiv-html: requested figures [7, 8] not found on https://arxiv.org/html/1609.00667 (available: 3, 4); ar5iv... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 74 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2507.18752.txt", "start": 25000, "length": 40000} | [chars 25000-61921 of 62265] (more available, next start=61921) zed by temperature cubed, $\eta\alpha$? n$\nu\alpha$-¯n$\nu\alpha$ T 3 = 2$\zeta$(3) $\pi$2 L$\alpha\approx$ L$\alpha$ 4 .(4) And, second, is the scaled lepton number density asym- metry normal... |
| 75 | hep-research-aid get_paper_figures | {"paper_id": "2507.20659", "figures": "1", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100"} | {"status": "error", "files": [], "message": "No usable HTML rendering for arXiv:2507.20659 (arxiv-html: https://arxiv.org/html/2507.20659 contains no extractable figures; ar5iv: https://ar5iv.labs.arx... |
| 76 | serve-sbbn.sh predict_sterile_nu_grid | {"masses_eV": [15000, 20000, 30000, 40000, 60000], "sin2_2thetas": [1e-15, 3e-15, 1e-14, 3e-14, 1e-13, 3e-13, 1e-12, 3e-12, 1e-11], "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-x... | { "status": "ok", "files": [ "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/sterile_nu_grid_DW_677e35.csv" ], "message": "Predicted 45 poi... |
| 77 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2507.18752.txt", "start": 50000, "length": 12300} | [chars 50000-62252 of 62265] (more available, next start=62252) r Astrophysics and Symmetries (N3AS), through the National Science Foundation Physics Frontier Center Award No. PHY-2020275. [1] A. Boy... |
| 78 | hep-research-aid read_text_file | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/sterile_nu_grid_DW_677e35.csv"} | m_s_eV,sin2_2theta, delta_neff,neff,omega_s_h2,Yp,DH_x1e5,Li7H_x1e10,chi2_bbn,delta_chi2_bbn,lifetime_s, free_streaming_kpc,bbn_excluded_95,overcloses, bbn_treatment_valid 15000,1e-15,1.1251e-08, 3.044,1.... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 79 | local read_arxiv_chunk | {"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2507.18752.txt", "length": 10000, "start": 36000} | [chars 36000-46000 of 62265] (more available, next start=46000) n the top panel of Fig. 5, so that the asym- metries are small enough that the structure formation is mostly sensitive to the sterile n... |
| 80 | serve-sbbn.sh predict_sterile_nu_xray_line | {"m_s_eV": 15000, "sin2_2theta": 1e-12} | { "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 15 keV, sin\u00b22\u03b8 = 1e-12\n\nE_\u03b3 = 7.5 keV\nDecay r... |
| 81 | serve-sbbn.sh predict_sterile_nu_xray_line | {"m_s_eV": 30000, "sin2_2theta": 1e-13} | { "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 30 keV, sin\u00b22\u03b8 = 1e-13\n\nE_\u03b3 = 15 keV\nDecay ra... |
| 82 | serve-sbbn.sh predict_sterile_nu_xray_line | {"m_s_eV": 40000, "sin2_2theta": 1e-14} | { "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 40 keV, sin\u00b22\u03b8 = 1e-14\n\nE_\u03b3 = 20 keV\nDecay ra... |
| 83 | serve-sbbn.sh predict_sterile_nu_xray_line | {"m_s_eV": 60000, "sin2_2theta": 1e-14} | { "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 60 keV, sin\u00b22\u03b8 = 1e-14\n\nE_\u03b3 = 30 keV\nDecay ra... |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 84 | local cite | {"arxiv_id": "2507.18752", "note": "[\"ticle massm th >8.5 keV (95% CL) from a combination\", \ "validated to remain within percentage-level error bounds\", \"the average error for\\nthe thermal mass fit was 1.3% while th... | QUOTE FAILED: "validated to remain within percentage-level error bounds" not found in source |
| 85 | local read_arxiv_chunk | {"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-sterile-bbn-xray/ runs/261021_143100/ 2510.24560.txt", "length": 30000, "start": 0} | [chars 0-30000 of 67438] (more available, next start=30000) Title: XRISM constraints on unidentified X-ray emission lines, including the 3.5 keV line, in the stacked spectrum of ten galaxy clusters A... |
| 86 | serve-sbbn.sh plot_sterile_nu_ bbn_impact | {"output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-sterile-bbn-xray/ runs/261021_143100"} | { "status": "ok", "files": [ "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-sterile-bbn-xray/ runs/261021_143100/ bbn_vs_neff.png" ], "message": "Plot saved to /Users/nramachan... [Fig. 12] |
| 87 | serve-sbbn.sh predict_sterile_ nu_grid | {"masses_eV": [7100, 50000] , "sin2_2thetas": [5e-09, 1e-08, 2.6e-08, 4.5e-08, 7.5e-08], "output_dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-sterile-bbn-xray/ runs/261021_143100"} | { "status": "ok", "files": [ "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-sterile-bbn-xray/ runs/261021_143100/ sterile_nu_grid_DW_ 7759e2.csv" ], "message": "Predicted 10 poi... |
| 88 | board post_finding | {"claim": "The tool grid (DW, m_s = 15, 20, 30, 40, 60 keV; sin^2 2theta = 1e-15 to 1e-11; 45 points) gives no BBN exclusion anywhere: \|Delta chi^2_ BBN\| <= 0.0044 and Delta N_ eff <= 3.4e-4. Only 60 keV at 1e-11 overclose... | Recorded finding f-015. |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 89 | board post_finding | {"claim": "Tool X-ray flags: (15 keV, 1e-12) is excluded by NuSTAR M31 (Roach+2020) and XMM blank sky (Dessert+2020), with Gamma = 1.03e-28 s^-1. (30 keV, 1e-13; Gamma = 3.31e-28), (40 keV, 1e-14; Gamma = 1.39e-28) and (... | Recorded finding f-016. |
| 90 | board post_dead_end | {"what": "get_paper_ figures on 1609.00667 (Figs. 7-8) and on 2507.20659 (Fig. 1), to digitize the NuSTAR limits and the zero-asymmetry lower edge", "why": "Neither paper has a usable HTML rendering. 1609.00667 exposes on... | Recorded dead end x-02. |
| 91 | serve-sbbn.sh predict_sterile_ nu_xray_line | {"m_s_eV": 20000, "sin2_ 2theta": 1e-13} | { "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 20 keV, sin\u00b22\u03b8 = 1e-13\n\nE_\u03b3 = 10 keV\nDecay ra... |
| 92 | serve-sbbn.sh predict_sterile_ nu_xray_line | {"m_s_eV": 30000, "sin2_ 2theta": 1e-11} | { "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 30 keV, sin\u00b22\u03b8 = 1e-11\n\nE_\u03b3 = 15 keV\nDecay ra... |
| 93 | board post_finding | {"claim": "In the sterile-nu tool, the X-ray exclusion boundary at m_ s = 20 keV lies between sin^2 2theta = 1e-13 (not excluded, Gamma = 4.36e-29 s^-1) and 1e-12 (excluded by NuSTAR M31, d-02). At 30 keV the tool flags n... | Recorded finding f-017. |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 94 | board post_question | {"text": "Can someone download the PDFs of 1609.00667 (Figs. 7-8) and 2507.20659 (Fig. 1) and read off the NuSTAR/ INTEGRAL upper edge and the zero-total-asymmetry lower edge at 15, 20, 30, 40 and 60 keV? HTML figure extr... | Recorded question q-06. |
| 95 | board post_finding | {"claim": "The m_th fit in 2507.18752 Eq. 8 has verbatim coefficients A = 1.329, a = 1.044, b = 0.055 (keV units), so at the paper's own benchmark (m_ s = 37 keV, sin^2 2theta = 1.5e-14) m_th = 1.329 x 37^ 1.044 x (1.5e-14... | Recorded finding f-018. |
| 96 | board post_finding | {"claim": "The consistent low-mass edge from m_th > 8.5 keV (Zelko+, 95% CL) uses the verified 2507.18752 Eq. 8: m_s,min = [8.5/ (1.329 (sin^2 2theta)^0.055) ]^(1/1.044) keV. By hand this gives m_s,min of about 36.5 keV at... | Recorded finding f-019. |
| 97 | board post_finding | {"claim": "I verified the f-001/f-002 items in 2507.18752 verbatim. The fit validity range is '15 keV≤m s≤60 keV and 10^ -15≤sin2(2θ)≤3×10^-13'; average errors are 1.3% (m_ th) and 8.3% (L). Further quotes: 'excluded at mu... | Recorded finding f-020. |
| 98 | board post_dead_end | {"what": "Verbatim re-extraction for f-004 (2405.06509 'generically unbound'), f-009 (XRISM 0.95e-6 ph/cm^2/s, Gamma < 0.97e-27 s^-1, f1(E) polynomial) and f-011 (1609.00667 flux formula, J_tot, DeltaOmega, m_s <~ 16 keV... | Recorded dead end x-03. |

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 99 | board post_finding | {"claim": "plot_sterile_nu_bbn_impact (Figure 12, bbn_vs_neff.png) gives an SM baseline of Yp = 0.24682 and D/H = 2.4433e-5, with slopes dYp/dN_eff = 0.01351 and d(D/H x1e5)/dN_eff = 0.3308 near the SM. One fully thermal... | Recorded finding f-021. |
| 100 | board post_finding | {"claim": "Correction to f-007 (my own reconstruction from the tool outputs): the tool's BBN chi^2 uses Yp and D/H only, not Planck N_eff. Its D/H error is about 0.065e-5 in total, which implies a D/H theory error of abo... | Recorded finding f-022. |
| 101 | board post_finding | {"claim": "The claim that BBN does not bind keV DW sterile-neutrino DM holds under every error budget I tested (own linearized estimates, using the tool slopes and the linear DW scaling Delta N_eff = 6.18e6 sin^2 2theta ... | Recorded finding f-023. |
| 102 | board post_finding | {"claim": "Own estimate, not a tool result: I used Delta Yp ~ -0.23 xi_e (literature-standard sensitivity) with xi ~ 1.46 L (asymmetry normalized to photons, T_nu = T_gamma before e+e- annihilation; this conversion depen... | Recorded finding f-024. |
| 103 | board post_question | {"text": "What residual xi_e at T ~ 1 MeV do the 2507.18752 (L_mu-dominated, L ~ 0.68 at 20 MeV) and 2507.20659 (L_e = -L_mu = 0.1) benchmarks actually reach after flavor equilibration, and is it below the ~0.03 needed t... | Recorded question q-07. |

### A.5 Files produced

- `scan_DW_9bfc3d.csv` (csv, scan_sterile_nu_parameters): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/scan_DW_9bfc3d.csv`
- `scan_DW_9bfc3d.npz` (other, scan_sterile_nu_parameters): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/scan_DW_9bfc3d.npz`
- `2405.06509.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2405.06509.txt`
- `2510.24560.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2510.24560.txt`
- `2507.18752.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2507.18752.txt`
- `sterile_nu_grid_DW_097459.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/sterile_nu_grid_DW_097459.csv`
- `1609.00667.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/1609.00667.txt`
- `2507.20659.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/2507.20659.txt`
- `sterile_nu_grid_DW_1b6ff9.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/sterile_nu_grid_DW_1b6ff9.csv`
- `sterile_nu_grid_DW_677e35.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/sterile_nu_grid_DW_677e35.csv`
- `sterile_nu_grid_DW_7759e2.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/sterile_nu_grid_DW_7759e2.csv`

## B Research board

The shared board of the research runs in this session, generated mechanically: what the team set out to investigate, what it found (with the tool calls that are the evidence), what failed, and how the critic judged each wave.

### B.1 Directions

| Id | Status | Goal | Outcome |
|---|---|---|---|
| d-01 | closed | Use the sterile-nu tools to compute BBN (Yp, D/H, N_eff) and relic-density constraints across the (m_s, sin^2 2theta) plane. Combine them with the tool's X-ray line predictions to map which regions remain viable. | The sterile-nu tools give BBN numbers only through Delta N_eff (a PRyMordial fit for Yp, D/H and Li7; exclusion at Delta chi^2 > 3.84 against Yp 0.2449 +/- 0.004, D/H 2.547 +/- 0.025 e-5 and Planck N_eff 2.99 +/- 0.17; valid below about 100 keV). Production is DW only in any reliable sense. The SF mode accepts a lepton_asymmetry but is a heuristic (0.1 x DW for Delta N_eff), and it returned an unp |
| d-02 | closed | Convert the 2024-2026 X-ray decay-rate limits (XRISM 2510.24560, with NuSTAR 1609.00667 as the high-mass reference) into sin^2 2theta(m_s) upper limits, and cross-check them against the tool's X-ray predictions. | I turned the XRISM 2510.24560 limits into upper limits on sin^2 2theta and checked the tool against them. One gap remains: I have no NuSTAR mixing-angle values at 10, 20 or 40 keV. 1609.00667 gives those limits only as curves in its Figs. 7-8, and I did not run get_paper_figures or digitize the plots before the budget ran out.<br>XRISM, cited verbatim: the 3-sigma limit on the decay rate at m_s = 7. |
| d-03 | closed | Extract the quantitative 2024-2026 resonant-production (lepton-asymmetry) results: the largest L allowed by BBN and the CMB, the lower bound on sin^2 2theta, and the Lyman-alpha mass cutoff. These define the lower and low-mass edges of the viable band. | Main result: the published resonant-production window survives only at large lepton asymmetry. It needs L >~ 0.5 (set at T = 10 GeV), m_s >~ 35 keV and sin^2 2theta <~ 1e-14 once X-ray limits and the structure bound m_th > 8.5 keV (95% CL) are applied. For L < 0.1 the window is excluded at much more than 95% CL. The sterile-neutrino tool cannot test this window.<br>There is no single largest L allow |
| d-04 | closed | Pin down the low-mass edge of the viable band from current structure-formation bounds (Lyman-alpha, Milky Way satellites, Tremaine-Gunn and phase-space limits), and map thermal-WDM mass bounds onto m_s for resonant spectra. | |

| Id | Status | Goal | Outcome |
|---|---|---|---|
| d-05 | blocked | Re-extract verbatim the numbers behind the contested findings f-001, f-002, f-004 and f-009. Resolve the m_th fit inconsistency so the low-mass edge of the resonant window is set consistently, absorbing the structure-formation edge goal of d-04. | The m_th discrepancy is resolved: 10.0 keV is correct and the 8.6 keV in f-002 was an arithmetic error. The other contested papers were not re-checked because the budget ran out. The f-001 and f-002 numbers in 2507.18752 are now confirmed verbatim (tc-79, tc-75). The m_th fit is Eq. 8: m_th = 1.329 m_s^1.044 (sin^2 2theta)^0.055, with masses in keV. The L fit is Eq. 9: L = 1.865e-12 m_s^-1.813 (si |
| d-06 | blocked | Quantify the upper (X-ray) and lower (asymmetry) edges of the viable band at 15, 20, 30, 40 and 60 keV, and assemble one viable-region table per asymmetry scenario. | I could not build the viable-region table, because the published X-ray edges and the zero-asymmetry lower edges are only available as plots I could not extract. What the tools did give is below.<br>1) Tool grid (DW production; m_s = 15, 20, 30, 40, 60 keV; sin^2 2theta from 1e-15 to 1e-11; 45 points; file /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100 |
| d-07 | closed | Robustness check of the BBN conclusions: test whether 'BBN does not bind keV sterile-neutrino DM' survives other error budgets, and estimate the Yp shift from lepton asymmetry that the tool omits. | 1. BBN impact plot (Figure 12, /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-sterile-bbn-xray/runs/261021_143100/bbn_vs_neff.png; tc-...-86). The plot depends only on N_eff, not on m_s, so no separate 7.1 keV run was needed. SM baseline: Yp = 0.24682, D/H = 2.4433e-5. Slopes near the SM: dYp/dN_eff = 0.01351 and d(D/H x1e5)/dN_eff = 0.3308. One thermalised sterile neutrino (N_eff = 4 |

## B.2 Findings and evidence

| Id | Confidence | Claim | Evidence (tool calls, figures) |
|---|---|---|---|
| f-001 | high, **contested** | Vogel et al. (2507.18752): Shi-Fuller sterile-nu DM is viable for initial L >~ 0.5 (L given at T=10 GeV), m_s >~ 35 keV and sin^2 2theta <~ 1e-14 once X-ray limits (NuSTAR+INTEGRAL/SPI, 95% CL) and the structure-formation bound m_th > 8.5 keV (Zelko+, 95% CL) are applied; the SF window is excluded at >>95% CL for L < 0.1; their Fig. 7 covers m_s 15-60 keV, sin^2 2theta 1e-15 to 1e-13. | tc-261021_1-w1-d-03-40, tc-261021_1-w1-d-03-35 |
| f-002 | high, **contested** | 2507.18752 provides fitting functions valid for 15 keV <= m_s <= 60 keV and 1e-15 <= sin^2 2theta <= 3e-13: m_th = 1.329 m_s^1.044 (sin^2 2theta)^0.055 (keV; average error 1.3%) and the L needed for all DM, L = 1.865e-12 m_s^-1.813 (sin^2 2theta)^-1.042 (average error 8.3%); also the mapping m_th,new = 0.83 m_th,old + 0.36 keV. Lyman-alpha (Villasenor+) prefers m_th = 4.1 (+33.3/-1.5) keV and the whole Fig. 7 region is within that 1-sigma range; galaxy counts give m_th > 8.4 keV, and counts plus lensing give > 8.5 keV (95% CL). My own evaluation of the fit, not a tool result: at the paper's be | tc-261021_1-w1-d-03-40, tc-261021_1-w1-d-03-35 |
| f-003 | high | Akita, Hamaguchi, Ovchynnikov (2507.20659): with large flavor asymmetries and zero total asymmetry (washed out by oscillations at T_osc ~ 15 MeV), the viable sin^2 2theta range grows by up to two orders of magnitude for m_s up to 60 keV. Individual asymmetries up to \|L_alpha\| ~ 0.1 are allowed by BBN/CMB (benchmarks L_e = -L_mu = 0.1 and L_mu = -L_tau = 0.06). The largest flavor-universal asymmetry allowed by BBN+CMB is L_e = L_mu = L_tau = 1e-3. Their Fig. 1 plane spans sin^2 2theta 1e-16 to 1e-8; the lower edge corresponds to sum L_alpha = 0. Lyman-alpha is shown only as an illustrative guid | tc-261021_1-w1-d-03-36, tc-261021_1-w1-d-03-39 |

| Id | Confidence | Claim | Evidence (tool calls, figures) |
|---|---|---|---|
| f-004 | high, **contested** | Model dependence: Froustey and Pitrou (2405.06509), using full 3-flavor neutrino transport plus BBN, find that 'the space of allowed initial asymmetries is generically unbound' with current abundance and CMB data, because flavor equilibration is not complete in general and the CMB bound is at least quadratic in the asymmetries. So there is no single L_max. The L >~ 0.5-1 (2507.18752) and \|L_alpha\| ~ 0.1 with sum = 0 (2507.20659) windows depend on the initial flavor configuration at T >~ 20 MeV. Only for flavor-universal asymmetry is the bound tight (about 1e-3). | tc-261021_1-w1-d-03-34, tc-261021_1-w1-d-03-36 |
| f-005 | high | Sterile-nu tools: BBN uses a PRyMordial fit (Yp, D/H, Li7) driven only by Delta N_eff, with exclusion at Delta chi^2 > 3.84 against Yp = 0.2449 +/- 0.004 (Aver+ 2021), D/H = 2.547 +/- 0.025 e-5 (Cooke+ 2018) and Planck N_eff = 2.99 +/- 0.17. The model is valid for m_s below about 100 keV. Production is DW; there is an SF option with a lepton_asymmetry parameter, but its Delta N_eff is a heuristic (0.1 x DW) and is not a resonant-production calculation. The tools do not model the effect of the asymmetry itself on BBN (xi_e). | tc-261021_1-w1-d-01-20, tc-261021_1-w1-d-01-48 |
| f-006 | high | DW scan (m_s 1-100 keV, sin^2 2theta 1e-14 to 1e-6, 60x60): 546 of 3600 points are BBN-excluded and 1740 overclose. The BBN 95% upper limit on sin^2 2theta is 3.9e-7 at 1 keV, 8.2e-8 at 7 keV, 4.4e-8 at 13 keV, 2.4e-8 at 26 keV, 1.7e-8 at 50 keV and 9.3e-9 at 100 keV. This is two or more orders of magnitude weaker than the overclosure boundary, so BBN does not constrain keV sterile-neutrino DM. | tc-261021_1-w1-d-01-25 |

| Id | Confidence | Claim | Evidence (tool calls, figures) |
|---|---|---|---|
| f-007 | high | Tool BBN numbers (DW, 84 points): the SM baseline is Yp = 0.246815, D/H = 2.4433e-5 and N_eff = 3.044. This is +0.48 sigma in Yp and -4.1 sigma in D/H from the observed values (chi^2 = 2.77). On the DW relic curve (Omega h^2 = 0.12: sin^2 2theta = 2.545e-10 at 7.1 keV, 1.374e-10 at 10 keV, 3.946e-11 at 20 keV, 7.583e-12 at 50 keV), Delta N_eff = 1.6e-3, 1.1e-3, 5.6e-4 and 2.2e-4, and Delta chi^2_BBN = -0.020, -0.014, -0.007 and -0.003. At 7.1 keV, sin^2 2theta = 3e-10 gives Yp = 0.246840 and D/H = 2.44395e-5. BBN starts to exclude only at sin^2 2theta around 1e-7 (7.1 keV: Delta N_eff = 0.62, | tc-261021_1-w1-d-01-31, tc-261021_1-w1-d-01-43 |
| f-008 | high | X-ray decay on the DW relic curve: Gamma = 6.25e-28 s^-1 (7.1 keV, E = 3.55 keV), 1.87e-27 (10 keV), 1.72e-26 (20 keV) and 3.23e-25 (50 keV). These agree within about 1.5% with the analytic 1.38e-29 (sin^2 2theta/1e-7)(m_s/keV)^5 formula (6.34e-28 at 7.1 keV, 3.27e-25 at 50 keV); the tool uses a 1.36e-29 prefactor. Tabulated limits exclude the DW-DM points at 7.1 and 10 keV (NuSTAR M31 Roach+2020 and XMM blank-sky Dessert+2020) and at 20 keV (NuSTAR M31). The 50 keV point is not excluded only because the tool has no tabulated limit within 30% in mass. Whole-halo Milky Way-center fluxes are 11. | tc-261021_1-w1-d-01-44, tc-261021_1-w1-d-01-45, tc-261021_1-w1-d-01-46, tc-261021_1-w1-d-01-47 |
| f-009 | high, **contested** | XRISM 2510.24560 (stacked spectra of 10 clusters, 3.75 Ms) gives a 3-sigma upper limit of 0.95e-6 ph/cm^2/s on line flux at the 3.55 keV line, which converts to a decay-rate limit of Gamma < 0.97e-27 s^-1 for m_s = 7.1 keV. The paper's mixing-angle fit, sin^2 2theta/1e-10 < 4.1 (E/3.55)^-4 f1(E), with f1 = 7.8774 - 3.5017E + 0.59087E^2 - 0.042245E^3 + 0.0010943E^4 (E in keV, valid 2.5-15 keV), evaluates by hand to the following upper limits on sin^2 2theta: about 1.3e-9 at m_s = 6 keV, about 4-5e-10 at 7.1 keV (3.9e-10 from Gamma; 4.8e-10 from the polynomial, f1 = 1.18), about 2.3e-10 at 8 keV | tc-261021_1-w1-d-02-37, tc-261021_1-w1-d-02-41 |

| Id | Confidence | Claim | Evidence (tool calls, figures) |
|---|---|---|---|
| f-010 | high | The tool's X-ray decay rate at m_s = 7.1 keV with sin^2 2theta = 4e-10 is 9.82e-28 s^-1 (lifetime 1.02e27 s). This agrees to within 2% with the XRISM limit of 0.97e-27 s^-1, so the tool's rate normalization matches the standard formula Gamma = 1.38e-32 s^-1 (sin^2 2theta/1e-10)(m/keV)^5 quoted in NuSTAR 1609.00667 Eq. 2. The tool also flags this point as excluded by NuSTAR M31 (Roach+ 2020) and XMM blank-sky (Dessert+ 2020), both stronger than XRISM. The tool's fluxes are whole-halo values (f_DM = 1, no field-of-view cut), so they cannot be compared directly with field-of-view-limited publishe | tc-261021_1-w1-d-02-30, tc-261021_1-w1-d-02-38, tc-261021_1-w1-d-02-41 |
| f-011 | high, **contested** | The NuSTAR Galactic Center paper (1609.00667) gives its sterile-neutrino line flux as F of about 2.6e-6 cm^-2 s^-1 (m/20 keV)^4 (sin^2 2theta/1e-14)(DeltaOmega/4 deg^2)(J/40 GeV cm^-3 kpc sr^-1), using the coreNFW default with J_tot = 29 and DeltaOmega_tot of about 3.8 deg^2. It states the limit improves on earlier ones by about an order of magnitude near m_s = 20 keV and implies m_s <~ 16 keV in the nuMSM. The paper gives its sin^2 2theta limits at 10, 20 and 40 keV only as curves in Figs. 7-8, not as numbers in the text, so the full-text extraction yields no verbatim values at those masses. | tc-261021_1-w1-d-02-38, tc-261021_1-w1-d-02-42 |
| f-012 | high | I cross-checked the tool against the 2507.18752 benchmark point, with SF production at m_s = 37 keV, sin^2 2theta = 1.5e-14 and L = 0.1 (the tool maximum). The tool gives Delta N_eff = 3.5e-8, Yp = 0.24682, D/H = 2.443e-5, Delta chi^2(BBN) = 0.00 (allowed), Gamma = 1.42e-28 s^-1 (E = 18.5 keV), and Omega_s h^2 = 25.1 (overcloses). The BBN output equals the SM value: the asymmetry does not enter the BBN abundances (no xi_e shift of n/p). Its SF Delta N_eff is a heuristic (0.1 x DW). The tool's SF relic density is about 200 times the observed DM, whereas the paper's fit puts this point at the co | tc-261021_1-w1-d-03-58, tc-261021_1-w1-d-03-40 |

| Id | Confidence | Claim | Evidence (tool calls, figures) |
|---|---|---|---|
| f-013 | medium | DW exclusion-plane plot for m_s 1-100 keV, sin^2 2theta 1e-14 to 1e-6 (Figure 10, exclusion_scan_DW_9bfc3d.png). Viable DW region: the Omega h^2 = 0.12 line runs from sin^2 2theta = 2.5e-10 at 7.1 keV to 7.6e-12 at 50 keV, with overclosure above it. BBN is irrelevant (it bites only above about 1e-8 to 1e-7). X-ray limits exclude the DW relic line for 7-20 keV. Below the line, sub-dominant DW sterile neutrinos are allowed, down to the X-ray limits. Full-DM viability needs resonant production, which the tool does not model reliably. | tc-261021_1-w1-d-01-57, tc-261021_1-w1-d-01-25, tc-261021_1-w1-d-01-43, tc-261021_1-w1-d-01-44, tc-261021_1-w1-d-01-46 Fig. 10 |
| f-014 | medium | Table of upper limits on sin^2 2theta from X-rays. From XRISM 2510.24560 (3-sigma limit, my arithmetic on the paper's polynomial): m_s = 6 keV, about 1.3e-9; 7.1 keV, about 4-5e-10 (3.9e-10 from Gamma < 0.97e-27 s^-1); 8 keV, about 2.3e-10; 10 keV, about 5.7e-11; 14 keV, about 1.2e-11. NuSTAR GC 1609.00667: improves on earlier limits by about 10x near 20 keV, but the numbers exist only in Figs. 7-8, so none are quoted. Tool check at m_s = 20 keV, sin^2 2theta = 1e-12: Gamma = 4.36e-28 s^-1, versus 4.42e-28 s^-1 from 1609.00667 Eq. 2 (agreement 1.4%). The tool flags this point as excluded by Nu | tc-261021_1-w1-d-02-41, tc-261021_1-w1-d-02-68, tc-261021_1-w1-d-02-38, tc-261021_1-w1-d-02-30, tc-261021_1-w1-d-02-66, ... |
| f-015 | high | The tool grid (DW, m_s = 15, 20, 30, 40, 60 keV; sin^2 2theta = 1e-15 to 1e-11; 45 points) gives no BBN exclusion anywhere: \|Delta chi^2_BBN\| <= 0.0044 and Delta N_eff <= 3.4e-4. Only 60 keV at 1e-11 overcloses under DW (Omega h^2 = 0.22). Lifetimes run from 9.68e27 s (15 keV, 1e-12) to 9.45e23 s (60 keV, 1e-11). The grid CSV has no X-ray exclusion column, so I took the X-ray flags from predict_sterile_nu_xray_line. | tc-261021_1-w2-d-06-76, tc-261021_1-w2-d-06-78 |

| Id | Confidence | Claim | Evidence (tool calls, figures) |
|---|---|---|---|
| f-016 | high, **contested** | Tool X-ray flags: (15 keV, 1e-12) is excluded by NuSTAR M31 (Roach+2020) and XMM blank sky (Dessert+2020), with Gamma = 1.03e-28 s^-1. (30 keV, 1e-13; Gamma = 3.31e-28), (40 keV, 1e-14; Gamma = 1.39e-28) and (60 keV, 1e-14; Gamma = 1.06e-27 s^-1) are all reported 'not excluded by tabulated limits (within 30% in mass)'. The tool's tabulated limits therefore do not reach the INTEGRAL/SPI or NuSTAR-GC regime above roughly 20-25 keV, so it cannot set the upper X-ray edge at 30-60 keV. | tc-261021_1-w2-d-06-80, tc-261021_1-w2-d-06-81, tc-261021_1-w2-d-06-82, tc-261021_1-w2-d-06-83 |
| f-017 | high, **contested** | In the sterile-nu tool, the X-ray exclusion boundary at m_s = 20 keV lies between sin^2 2theta = 1e-13 (not excluded, Gamma = 4.36e-29 s^-1) and 1e-12 (excluded by NuSTAR M31, d-02). At 30 keV the tool flags nothing even at sin^2 2theta = 1e-11 (Gamma = 3.31e-26 s^-1, E = 15 keV). The tool therefore has no tabulated X-ray limit for line energies above about 10-13 keV (m_s >~ 25 keV), which explains why it cannot reproduce the NuSTAR-GC / INTEGRAL upper edge at 30-60 keV. | tc-261021_1-w2-d-06-91, tc-261021_1-w2-d-06-92, tc-261021_1-w2-d-06-81, tc-261021_1-w2-d-06-83 |
| f-018 | high, **contested** | The m_th fit in 2507.18752 Eq. 8 has verbatim coefficients A = 1.329, a = 1.044, b = 0.055 (keV units), so at the paper's own benchmark (m_s = 37 keV, sin^2 2theta = 1.5e-14) m_th = 1.329 x 37^1.044 x (1.5e-14)^0.055 = 10.0 keV, not 8.6 keV. The f-002 value of 8.6 keV is an arithmetic error. The L fit (Eq. 9: C = 1.865e-12, alpha = -1.813, beta = -1.042) gives L = 0.68 at the same point, which confirms that part of f-002. | tc-261021_1-w2-d-05-79 |
| f-019 | medium, **contested** | The consistent low-mass edge from m_th > 8.5 keV (Zelko+, 95% CL) uses the verified 2507.18752 Eq. 8: m_s,min = [8.5/(1.329 (sin^2 2theta)^0.055)]^(1/1.044) keV. By hand this gives m_s,min of about 36.5 keV at sin^2 2theta = 1e-15, 32.3 keV at 1e-14, 31.6 keV at 1.5e-14 and 28.6 keV at 1e-13. The L = 0.5 contour (Eq. 9) crosses the m_th = 8.5 keV line near m_s of about 31 keV and sin^2 2theta of about 2.8e-14. At sin^2 2theta = 1e-14 the edge point needs L of about 1.35. The paper's rounded summary 'm_s >~ 35 keV, sin^2 2theta <~ 1e-14, L >~ 0.5' (verbatim, Sec. IV) is consistent with this to | tc-261021_1-w2-d-05-79 |

| Id | Confidence | Claim | Evidence (tool calls, figures) |
|---|---|---|---|
| f-020 | high, **contested** | I verified the f-001/f-002 items in 2507.18752 verbatim. The fit validity range is '15 keV≤m s≤60 keV and 10^-15≤sin2(2$\theta$)≤3×10^-13'; average errors are 1.3% (m_th) and 8.3% (L). Further quotes: 'excluded at much greater than 95% CL forL<0.1'; 'mth = 4.1+33.3 -1.5 keV [20]'; 'mth >8.4 keV (95% CL) [18] andm th >8.5 keV (95% CL) [19]'; Eq. 7 'mth,new≈0.83m th,old + 0.36'; Fig. 7 caption: initial L 'given atT= 10 GeV' and 'reduced by a factor of 5 to 10 byT= 10 MeV'. Fig. 7 has m_s axis ticks 15-60 keV and sin^2 2theta tick labels 1e-15, 1e-14 and 1e-13. This does not contradict the fit-validity | tc-261021_1-w2-d-05-79, tc-261021_1-w2-d-05-75 |
| f-021 | high | plot_sterile_nu_bbn_impact (Figure 12, bbn_vs_neff.png) gives an SM baseline of Yp = 0.24682 and D/H = 2.4433e-5, with slopes dYp/dN_eff = 0.01351 and d(D/H x1e5)/dN_eff = 0.3308 near the SM. One fully thermalised sterile neutrino (N_eff = 4.044) gives Yp = 0.2596 and D/H = 2.777e-5. The tool's BBN-only limit is Delta N_eff < 0.548 (95%). The plot does not depend on m_s: it is a function of N_eff only, so there is no separate 7.1 keV run. | tc-261021_1-w2-d-07-86 Fig. 12 |
| f-022 | medium | Correction to f-007 (my own reconstruction from the tool outputs): the tool's BBN chi^2 uses Yp and D/H only, not Planck N_eff. Its D/H error is about 0.065e-5 in total, which implies a D/H theory error of about 0.060e-5 added in quadrature to the observational 0.025e-5. With that error the SM D/H offset is -1.6 sigma, not -4.1 sigma; -4.1 sigma holds only with the observational error alone. Check: baseline chi^2 = 0.23 (Yp) + 2.54 (D/H) = 2.77. This reproduces the tool's Delta chi^2 = -1.14 at Delta N_eff = 0.161, Delta chi^2 = 5.9 at Delta N_eff = 0.62, and the Delta N_eff < 0.548 limit. | tc-261021_1-w2-d-07-86, tc-261021_1-w2-d-07-87 |

| Id | Confidence | Claim | Evidence (tool calls, figures) |
|---|---|---|---|
| f-023 | medium | The claim that BBN does not bind keV DW sterile-neutrino DM holds under every error budget I tested (own linearized estimates, using the tool slopes and the linear DW scaling Delta N_eff = 6.18e6 sin^2 2theta at 7.1 keV and 2.95e7 sin^2 2theta at 50 keV taken from the grid). The 95% limits are: - Yp only: Delta N_eff < 0.455 - Tool default (Yp plus D/H with theory error): < 0.548 - Yp plus D/H with observational error only (SM chi^2 = 17.4): < 0.61 - D/H only with theory error: < 0.81 - Planck N_eff 2.99 +/- 0.17, as an extreme check (a keV sterile neutrino is non-relativistic at recombination | tc-261021_1-w2-d-07-86, tc-261021_1-w2-d-07-87 |
| f-024 | low, **contested** | Own estimate, not a tool result: I used Delta Yp ~ -0.23 xi_e (literature-standard sensitivity) with xi ~ 1.46 L (asymmetry normalized to photons, T_nu = T_gamma before e+e- annihilation; this conversion depends on convention at the factor-of-4 level). - Flavor-universal L = 1e-3 gives xi_e ~ 1.5e-3 and Delta Yp ~ -3.4e-4, or 0.08 sigma against sigma_Yp = 0.004. Negligible. - A 1 sigma (2 sigma) Yp shift needs a residual xi_e ~ 0.017 (0.035), i.e. a residual L_e ~ 0.012 (0.024). - 2507.20659 (L_e = -L_mu = 0.1): without equilibration, xi_e ~ 0.15 and Delta Yp ~ -0.034 (about 8 sigma). Their wi | tc-261021_1-w2-d-07-86 |

### B.3 Dead ends

- x-01 (d-01): Using the tool's SF (Shi-Fuller) mode with lepton_asymmetry to compute resonant relic density and BBN. — The tool labels SF Delta N_eff as a heuristic (0.1 x DW). It returned Omega h^2 = 432 at 7.1 keV, sin^2 2theta = 7e-11, L = 1e-3, which is unphysical. BBN does not include the effect of the lepton asymmetry itself (xi_e on Yp). Resonant-band viability must come from the literature (d-03).

- x-02 (d-06): get_paper_figures on 1609.00667 (Figs. 7-8) and on 2507.20659 (Fig. 1), to digitize the NuSTAR limits and the zero-asymmetry lower edge — Neither paper has a usable HTML rendering. 1609.00667 exposes only Figs. 3-4, and 2507.20659 has no extractable figures. A PDF download plus manual inspection would be needed, and the tools cannot digitize a plot.

- x-03 (d-05): Verbatim re-extraction for f-004 (2405.06509 'generically unbound'), f-009 (XRISM 0.95e-6 ph/cm^2/s, Gamma < 0.97e-27 s^-1, f1(E) polynomial) and f-011 (1609.00667 flux formula, J_tot, DeltaOmega, m_s <~ 16 keV), plus the cite() of 2507.18752 quotes. — The budget ran out. The XRISM chunk 0-30000 contained only the author and affiliation list. 2405.06509 and 1609.00667 were not read. The cite() call failed because

one quote ('validated to remain within percentage-level error bounds') does not match the extracted text, which has line breaks. These values stay unverified in this direction ('not found', not 'refuted'). The next attempt should read 2

### B.4 Critic verdicts

- `c-01` (wave 0, plan): **revise**; high: Step 1 must check the tool's validity range and production mechanism before the scan is configured. If the tool models only Dodelson-Widrow production, a 1e-14 to 1e-6 grid is mostly wasted, and the resonant window cannot be called tool-computed. Make the scan bounds depend on what describe_sterile_; high: A keV-mass sterile neutrino that is non-relativistic at BBN and has tiny mixing gives Delta N_eff of about 0, so Yp and D/H will sit at their Standard Model values. Do not present that as a BBN constraint on the plane. State whether the tool's BBN output responds to m_s and sin^2 2theta at all, and ; medium: No general plotting tool exists, so a combined literature-plus-tool overlay cannot be produced. Report only the plot_sterile_nu_exclusion and plot_sterile_nu_bbn_impact paths, and give the combined viable band as a table. check_sterile_nu_sbl_anomaly is off-task and should be dropped if the budget i; high: The brief's X-ray picture is outdated or incomplete. NuSTAR limits newer than 1609.00667 (Roach et al. 2022-2023, M31 and Galactic bulge) likely supersede it at 6-40 keV. XMM blank-sky limits (Dessert et al., 1812.06976) are probably stronger than XRISM at 3.5 keV. INTEGRAL/SPI should cover m_s abov; medium: The brief's XRISM numbers (1.0e-27 s^-1, 3-4 times Hitomi, about 5 times weaker than the XMM rate) are unverified. Quote them verbatim or drop them. The analytic conversion is internally consistent (1e-27 s^-1 gives about 4e-10, and 7e-11 gives about 1.7e-28 s^-1). The factor-of-2 tool match also ne; high: Over budget. Four papers at download, read chunks and cite each, plus figures, comes to more than 13 calls. Prioritize 2507.20659 (maximal mixing range) and 2405.06509 (L_max, Yp, D/H). Use abstracts or figures only for 2507.18752 and 2502.17374.

- `c-02` (wave 1, wave): **continue**; contested `f-001`, `f-002`, `f-004`, `f-009`, `f-011`; medium: The baseline numbers are internally inconsistent. Offsets of +0.48 sigma in Yp and -4.15 sigma in D/H (sigma = 0.025e-5) give chi^2 of about 17.4, not the tool's 2.77. The tool's likelihood must include an extra D/H theory error, so the '-4.1 sigma' statement misdescribes it. The negative Delta chi^; low: The 1.5% gap from the analytic formula comes mostly from display rounding (2.5e-10 printed, 2.545e-10 used). The missing 50 keV limit is a gap in the tool, not a physical allowance. Published NuSTAR GC, INTEGRAL and Lyman-alpha limits exclude DW-DM at 50 keV, and the write-up should say so.; medium: The DW viable region relies on the tool's tabulated X-ray limits (Roach+ 2020, Dessert+ 2020). Their numerical sin^2 2theta curves were never extracted, so 'below the X-ray limits' is not quantified. The XMM blank-sky limit itself is contested in the literature (Boyarsky et al. vs Dessert et al.), a; medium: There are no numerical X-ray limits above 14 keV. The NuSTAR and INTEGRAL curves and the newer NuSTAR M31, Galactic bulge and eROSITA results were not digitized. The upper edge of the resonant window at 15-60 keV is therefore unquantified, and the final viable-region answer needs it.; high: The viable band is not yet assembled. The lower edge in sin^2 2theta at fixed masses from 2507.20659 and 2502.17374 was not extracted. The m_s >~ 35 keV edge depends on an m_th fit that does not reproduce on re-evaluation. The literature claims behind f-001, f-002, f-004 and f-009 still need verbati

- `c-03` (wave 2, wave): **continue**; contested `f-016`, `f-017`, `f-018`, `f-019`, `f-020`, `f-024`; high: The board brief still states 'viable regions at m_s >~ 10 keV' and 'no such tools were available to me'. Both are now wrong: the low-mass edge is about 30-37 keV (paper: >~ 35

keV), and the sterile-nu tools were used extensively. The final synthesis must correct the brief.; medium: The m_th convention is unchecked. Eq. 7 (m_th,new = 0.83 m_th,old + 0.36) shows the paper distinguishes two conventions. The low-mass edge moves by several keV depending on which convention the 8.5 keV bound and the Eq. 8 output use.; high: There is still no quantitative X-ray upper edge at m_s = 30-60 keV. This is exactly the mass range where the only viable full-DM window (f-001) sits, so the central question of which region is viable is answered only by quoting 2507.18752's rounded summary, not by an independent cross-check.; medium: f-001, f-002, f-004, f-009 and f-011 remain contested across waves because of missing verbatim excerpts. references.bib lacks the 2507.18752 quotes because cite() failed on a line-broken string, and the tc-42 misattribution is still uncorrected.; low: f-022 says the tool chi^2 excludes Planck N_eff, which contradicts the d-01 description of the tool's criterion. tc-87 supports f-022: at Delta N_eff = 0.031 the tool's Delta chi^2 = -0.367 matches Yp plus D/H only (reconstruction gives -0.36; adding Planck would give about -0.21). The d-01 descript

- `c-04` (wave 2, report): **revise**; high: unsupported: **DW production:** only a sub-dominant component, below the relic line and below the X-ray — The report stops mid-sentence in the Discussion. The 'What survives' synthesis is missing, and that is the answer to the task's central question of which regions remain viable. The board's per-s; high: unsupported: it depends on an unchecked m_th convention (see Discussion) — The Discussion never addresses the m_th convention. Critique c-03 flags it: the Eq. 7 old/new mapping can shift the low-mass edge by several keV. Because the report is truncated, this caveat never appears, even though the tex